Supporting Document
Mandatory Technical Document

PP-Module for IPS
Version: 2.0
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National Information Assurance Partnership

Foreword

This is a Supporting Document (SD), intended to complement the Common Criteria version 3 and the associated Common Evaluation Methodology for Information Technology Security Evaluation.

SDs may be “Guidance Documents”, that highlight specific approaches and application of the standard to areas where no mutual recognition of its application is required, and as such, are not of normative nature, or “Mandatory Technical Documents”, whose application is mandatory for evaluations whose scope is covered by that of the SD. The usage of the latter class is not only mandatory, but certificates issued as a result of their application are recognized under the CCRA.

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Document history:

Version	Date	Comment
1.0	2021-05-11	Initial publication
2.0	2026-03-10	CC:2022 updates, incorporation of TDs

General Purpose:
The purpose of this SD is to define evaluation methods for the functional behavior of IPS products.

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1 Introduction

1.1 Technology Area and Scope of Supporting Document

The scope of the PP-Module for IPS is to describe the security functionality of IPS products in terms of [CC] and to define functional and assurance requirements for them. The PP-Module is intended for use with the following Base-PP:

- Network Device, version 4.0.

This SD is mandatory for evaluations of TOEs that claim conformance to a PP-Configuration that includes the PP-Module for :

- IPS, Version 2.0

As such it defines Evaluation Activities for the functionality described in the PP-Module as well as any impacts to the Evaluation Activities to the Base-PP(s) it modifies.

Although Evaluation Activities are defined mainly for the evaluators to follow, in general they also help developers to prepare for evaluation by identifying specific requirements for their TOE. The specific requirements in Evaluation Activities may in some cases clarify the meaning of Security Functional Requirements (SFR), and may identify particular requirements for the content of Security Targets (ST) (especially the TOE Summary Specification), user guidance documentation, and possibly supplementary information (e.g. for entropy analysis or cryptographic key management architecture).

1.2 Structure of the Document

Evaluation Activities can be defined for both SFRs and Security Assurance Requirements (SAR), which are themselves defined in separate sections of the SD.

If any Evaluation Activity cannot be successfully completed in an evaluation, then the overall verdict for the evaluation is a 'fail'. In rare cases there may be acceptable reasons why an Evaluation Activity may be modified or deemed not applicable for a particular TOE, but this must be approved by the Certification Body for the evaluation.

In general, if all Evaluation Activities (for both SFRs and SARs) are successfully completed in an evaluation then it would be expected that the overall verdict for the evaluation is a ‘pass’. To reach a ‘fail’ verdict when the Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

Similarly, at the more granular level of assurance components, if the Evaluation Activities for an assurance component and all of its related SFR Evaluation Activities are successfully completed in an evaluation then it would be expected that the verdict for the assurance component is a ‘pass’. To reach a ‘fail’ verdict for the assurance component when these Evaluation Activities have been successfully completed would require a specific justification from the evaluator as to why the Evaluation Activities were not sufficient for that TOE.

1.3 Terms

The following sections list Common Criteria and technology terms used in this document.

1.3.1 Common Criteria Terms

Assurance	Grounds for confidence that a TOE meets the SFRs [CC].
Base Protection Profile (Base-PP)	Protection Profile used as a basis to build a PP-Configuration.
Collaborative Protection Profile (cPP)	A Protection Profile developed by international technical communities and approved by multiple schemes.
Common Criteria (CC)	Common Criteria for Information Technology Security Evaluation (International Standard ISO/IEC 15408).
Common Criteria Testing Laboratory	Within the context of the Common Criteria Evaluation and Validation Scheme (CCEVS), an IT security evaluation facility accredited by the National Voluntary Laboratory Accreditation Program (NVLAP) and approved by the NIAP Validation Body to conduct Common Criteria-based evaluations.
Common Evaluation Methodology (CEM)	Common Evaluation Methodology for Information Technology Security Evaluation.

Distributed TOE	A TOE composed of multiple components operating as a logical whole.
Functional Package (FP)	A document that collects SFRs for a particular protocol, technology, or functionality.
Operational Environment (OE)	Hardware and software that are outside the TOE boundary that support the TOE functionality and security policy.
Protection Profile (PP)	An implementation-independent set of security requirements for a category of products.
Protection Profile Configuration (PP-Configuration)	A comprehensive set of security requirements for a product type that consists of at least one Base-PP and at least one PP-Module.
Protection Profile Module (PP-Module)	An implementation-independent statement of security needs for a TOE type complementary to one or more Base-PPs.
Security Assurance Requirement (SAR)	A requirement to assure the security of the TOE.
Security Functional Requirement (SFR)	A requirement for security enforcement by the TOE.
Security Target (ST)	A set of implementation-dependent security requirements for a specific product.
Target of Evaluation (TOE)	The product under evaluation.
TOE Security Functionality (TSF)	The security functionality of the product under evaluation.
TOE Summary Specification (TSS)	A description of how a TOE satisfies the SFRs in an ST.

1.3.2 Technical Terms

Anomaly / Anomalous (network traffic)	Traffic that does not fit into a defined baseline and is therefore unexpected or atypical traffic. Anomalous traffic is not necessarily dangerous, and does not necessarily indicate any threat to the monitored network.
Baseline / Base-lining (network traffic)	Defining what is to be considered expected or typical network traffic on a monitored network. A traffic baseline does not indicate that all traffic that matches the baseline is safe, or that the traffic is not a potential threat to the monitored network. For example: traffic that matches a baseline can still match a list of known-bad IP addresses; or can match signatures of known threats.
Flooding	Causing an excessive amount of traffic on an IP subnet or targeted against a specific IP address.
IPS policy	Any set of rules for traffic analysis, traffic blocking, signature detection, and/or anomaly detection. Many IPS policies could be defined and stored on the TOE, but an IPS policy will not have any affect unless is applied to (made active on) one or more IPS interfaces.

Inline mode	The deployment of the TOE (or TOE component) such that monitored network traffic must flow across the TOE, thus providing the TOE with the opportunity to block the traffic.
Normalization (of network traffic)	Filtering of network traffic such that only the useful packets/fragments are allowed through to the destination. Normalization can only be performed by the TOE when the TOE is deployed in inline mode. Normalization can include filtering out any type of 'known-good' traffic that is expected to be present on the network and therefore not considered to be an indicator of potential intrusion.
Profiling (network traffic)	See base-lining.
Promiscuous mode	The state of an IPS interface in which it is collecting and inspecting network traffic. A promiscuous interface could be one that is only listening and never transmitting traffic, or could be an interface through which traffic flows both inbound and outbound as in an inline mode deployment.
Sensor interface	Any interface of the TOE that has an IPS policy applied to it.

2 Evaluation Activities for SFRs

The EAs presented in this section capture the actions the evaluator performs to address technology specific aspects covering specific SARs (e.g. ASE_TSS.1, ADV_FSP.1, AGD_OPE.1, and ATE_IND.1) – this is in addition to the CEM workunits that are performed in Section 3 Evaluation Activities for SARs.

Regarding design descriptions (designated by the subsections labeled TSS, as well as any required supplementary material that may be treated as proprietary), the evaluator must ensure there is specific information that satisfies the EA. For findings regarding the TSS section, the evaluator's verdicts will be associated with the CEM workunit ASE_TSS.1-1. Evaluator verdicts associated with the supplementary evidence will also be associated with ASE_TSS.1-1, since the requirement to provide such evidence is specified in ASE in the PP.

For ensuring the guidance documentation provides sufficient information for the administrators/users as it pertains to SFRs, the evaluator's verdicts will be associated with CEM workunits ADV_FSP.1-7, AGD_OPE.1-4, and AGD_OPE.1-5.

Finally, the subsection labeled Tests is where the authors have determined that testing of the product in the context of the associated SFR is necessary. While the evaluator is expected to develop tests, there may be instances where it is more practical for the developer to construct tests, or where the developer may have existing tests. Therefore, it is acceptable for the evaluator to witness developer-generated tests in lieu of executing the tests. In this case, the evaluator must ensure the developer's tests are executing both in the manner declared by the developer and as mandated by the EA. The CEM workunits that are associated with the EAs specified in this section are: ATE_IND.1-3, ATE_IND.1-4, ATE_IND.1-5, ATE_IND.1-6, and ATE_IND.1-7.

2.1 Network Device

The EAs defined in this section are only applicable in cases where the TOE claims conformance to a PP-Configuration that includes the NDcPP.

2.1.1 Modified SFRs

The PP-Module does not modify any requirements when the NDcPP is the base.

2.2 TOE SFR Evaluation Activities

2.2.1 Security Audit (FAU)

FAU_GEN.1/IPS Audit Data Generation (IPS)

FAU_GEN.1/IPS
TSS

The evaluator shall verify that the TSS describes how the TOE can be configured to log IPS data associated with applicable policies.

The evaluator shall verify that the TSS describes what (similar) IPS event types the TOE will combine into a single audit record along with the conditions (e.g., thresholds and time periods) for so doing. The TSS shall also describe to what extent (if any) that may be configurable.

For FIP_SBD_EXT.1, for each field, the evaluator shall verify that the TSS describes how the field is inspected and if logging is not applicable, any other mechanism such as counting that is deployed.

Guidance
The evaluator shall verify that the operational guidance describes how to configure the TOE to result in applicable IPS data logging.
Tests

The evaluator shall test that the interfaces used to configure the IPS polices yield expected IPS data in association with the IPS policies. A number of IPS policy combination and ordering scenarios need to be configured and tested by attempting to pass both allowed and anomalous network traffic matching configured IPS policies in order to trigger all required IPS events.

Note the following:

- This testing is addressed with a combination of the Test EAs for the other IPS requirements
- As part of testing this activity, the evaluator shall also ensure that the audit data the TSF generates to conform to this SFR can be transmitted to an external audit server in the manner that the Base-PP requires for all audit data.

2.2.2 Intrusion Prevention (FIP)

FIP_ABD_EXT.1 Anomaly-Based IPS Functionality

FIP_ABD_EXT.1
TSS

The evaluator shall verify that the TSS describes the composition, construction, and application of baselines or anomaly-based attributes specified in FIP_ABD_EXT.1.1.

If ‘frequency’ is selected in FIP_ABD_EXT.1.1, the TSS shall include an explanation of how frequencies can be defined on the TOE.

If ‘thresholds’ is selected in FIP_ABD_EXT.1.1, the TSS shall include an explanation of how the thresholds can be defined on the TOE.

The evaluator shall verify that each baseline or anomaly-based rule can be associated with a reaction specified in FIP_ABD_EXT.1.3.

The evaluator shall verify that the TSS identifies all interface types capable of applying baseline or anomaly-based rules and explains how they are associated with distinct network interfaces. Where interfaces can be grouped into a common interface type (e.g., where the same internal logical path is used, perhaps where a common device driver is used) they can be treated collectively as a distinct network interface.

Guidance

The evaluator shall verify that the operational guidance provides instructions to manually create baselines or anomaly-based rules according to the selections made in FIP_ABD_EXT.1.1. Note that dynamic “profiling” of a network to establish a baseline is outside the scope of the PP-Module

The evaluator shall verify that the operational guidance provides instructions to associate reactions specified in FIP_ABD_EXT.1.3 with baselines or anomaly-based rules.

The evaluator shall verify that the operational guidance provides instructions to associate the different policies with distinct network interfaces.

Tests

The evaluator shall perform the following tests:

- Test FIP_ABD_EXT.1:1: The evaluator shall use the instructions in the operational guidance to configure baselines or anomaly-based rules for each attribute specified in FIP_ABD_EXT.1.1. The evaluator shall send traffic that does not match the baseline or matches the anomaly-based rule and verify the TOE applies the configured reaction. This shall be performed for each attribute in FIP_ABD_EXT.1.1.
- Test FIP_ABD_EXT.1:2: The evaluator shall repeat the test above to ensure that baselines or anomaly-based rules can be defined for each distinct network interface type supported by the TOE.

FIP_IPB_EXT.1 IP Blocking

FIP_IPB_EXT.1

TSS

The evaluator shall verify how good/bad lists affect the way in which traffic is analyzed with respect to processing packets. The evaluator shall also very that the TSS provides details for the attributes that create a known good list, a known bad list, and their associated rules, including how to define the source or destination IP address (e.g. a single IP address or a range of IP addresses).

If the TSF uses address types other than a single IP or a range of IP addresses (e.g., MAC addresses), the evaluator shall check that the TSS explains what configurations would cause non-IP lists of known-good and known-bad addresses to take precedence over IP-based address lists.

Guidance

The evaluator shall verify that the administrative guidance provides instructions with how each role specified in the requirement can create, modify and delete the attributes of known-good and known-bad lists.

If the TSF uses address types other than a single IP or a range of IP addresses (e.g., MAC addresses), the evaluator shall check that the operational guidance includes instructions for any configurations that would cause non-IP lists of known-good and known-bad addresses to take precedence over IP-based address lists.

Tests

The evaluator shall perform the following tests:

- Test FIP_IPB_EXT.1:1: The evaluator shall use the instructions in the operational guidance to create a known-bad address list. Using a single IP address, a list of addresses, or a range of addresses from that list, the evaluator shall attempt to send traffic through the TOE that would otherwise be allowed by the TOE and observe the TOE automatically drops that traffic.
- Test FIP_IPB_EXT.1:2: The evaluator shall use the instructions in the operational guidance to create a known-good address list. Using a single IP address, a list of addresses, or a range of addresses from that list, the evaluator shall attempt to send traffic that would otherwise be denied by the TOE and observe the TOE automatically allowing traffic.
- Test FIP_IPB_EXT.1:3: The evaluator shall add conflicting IP addresses to each list and ensure that the TOE handles conflicting traffic in a manner consistent with the precedence in FIP_NTA_EXT.1.1.

FIP_NTA_EXT.1 Network Traffic Analysis

FIP_NTA_EXT.1.1
TSS

The evaluator shall verify that the TSS explains the TOE's capability of analyzing IP traffic in terms of the TOE's policy hierarchy (precedence). The TSS should identify if the TOE's policy hierarchy order is configurable by the administrator for IPS policy elements (known-good lists, known-bad lists, signature-based rules, and anomaly-based rules).

Regardless of whether the precedence is configurable, the evaluator shall verify that the TSS describes the default precedence as well as the IP analyzing functions supported by the TOE.

Guidance

The evaluator shall verify that the guidance describes the default precedence.

If the precedence is configurable, the evaluator shall verify that the guidance explains how to configure the precedence.

Tests
There are no test EAs for this element.
FIP_NTA_EXT.1.2
TSS
The evaluator shall verify that the TSS indicates that the following protocols are supported:

- IPv4
- IPv6
- ICMPv4
- ICMPv6
- TCP
- UDP

The evaluator shall verify that the TSS describes how conformance with the identified protocols has been determined by the TOE developer (e.g., third party interoperability testing, protocol compliance testing).
Guidance
There are no guidance EAs for this element.
Tests
There are no test EAs for this element.
FIP_NTA_EXT.1.3
TSS

The evaluator shall verify that the TSS identifies all interface types capable of being deployed in the modes of promiscuous, and or inline mode as well as the interfaces necessary to facilitate each deployment mode (at a minimum, the interfaces need to support inline mode). The evaluator shall also check that the TSS provides a description for how the management interface is logically distinct from any sensor interfaces.
Guidance

The evaluator shall verify that the operational guidance provides instructions on how to deploy each of the deployment methods outlined in the TSS. The evaluator shall also verify that the operational guidance provides instructions of applying IPS policies to interfaces for each deployment mode. If the management interface is configurable, the evaluator shall verify that the operational guidance explains how to configure the interface as a management interface.

The evaluator shall verify that the operational guidance explains how the TOE sends commands to remote traffic filtering devices if this functionality is supported.

Tests
Testing for this element is performed in conjunction with testing where promiscuous and inline interfaces are tested.

FIP_SBD_EXT.1 Signature-Based IPS Functionality

FIP_SBD_EXT.1.1
TSS

The evaluator shall verify that the TSS describes what is comprised within a signature rule.

The evaluator shall verify that each signature can be associated with a reaction specified in FIP_SBD_EXT.1.5.

The evaluator shall verify that the TSS identifies all interface types that are capable of applying signatures and explains how rules are associated with distinct network interfaces. Where interfaces can be grouped into a common interface type (e.g., where the same internal logical path is used, perhaps where a common device driver is used) they can be treated collectively as a distinct network interface.

Guidance
The evaluator shall verify that the operational guidance provides instructions with how to create and/or configure rules using the following protocols and header inspection fields:

- IPv4: version; header length; packet length; ID; IP flags; fragment offset; time to live (TTL); protocol; header checksum; source address; destination address; IP options; and, if selected, type of service (ToS).
- IPv6: Version; payload length; next header; hop limit; source address; destination address; routing header; and, if selected, traffic class and/or flow label
- ICMP: type; code; header checksum; and, if selected, other header fields (varies based on the ICMP type and code).
- ICMPv6: type; code; and header checksum.
- TCP: Source port; destination port; sequence number; acknowledgment number; offset; reserved; TCP flags; window; checksum; urgent pointer; and TCP options.
- UDP: source port; destination port; length; and UDP checksum.

The evaluator shall verify that the operational guidance provides instructions with how to select and/or configure reactions specified in FIP_SBD_EXT.1.5 in the signature rules.
Tests
The evaluator shall perform the following tests:

- Test FIP_SBD_EXT.1.1:1: The evaluator shall use the instructions in the operational guidance to test that packet header signatures can be created and/or configured with the selected and/or configured reactions

specified in FIP_SBD_EXT.1.5 for each of the attributes listed below. Each attribute shall be individually assigned to its own unique signature:
    - IPv4: version; header length; packet length; ID; IP flags; fragment offset; time to live (TTL); protocol; header checksum; source address; destination address; IP options; and, if selected, type of service (ToS).
    - IPv6: Version; payload length; next header; hop limit; source address; destination address; routing header; and, if selected, traffic class and/or flow label
    - ICMP: type; code; header checksum; and, if selected, other header fields (varies based on the ICMP type and code).
    - ICMPv6: Type; Code; and Header Checksum.
    - TCP: Source port; destination port; sequence number; acknowledgment number; offset; reserved; TCP flags; window; checksum; urgent pointer; and TCP options.
    - UDP: Source port; destination port; length; and UDP checksum.

  The evaluator shall generate traffic to trigger a signature and shall then use a packet sniffer to capture traffic that ensures the reactions of each rule are performed as expected.
- Test FIP_SBD_EXT.1.1:2: The evaluator shall repeat the test above to ensure that signature-based IPS policies can be defined for each distinct network interface type capable of applying signatures as supported by the TOE.

FIP_SBD_EXT.1.2

TSS

The evaluator shall verify that the TSS describes what is comprised within a string-based detection signature.

The evaluator shall verify that each packet payload string-based detection signature can be associated with a reaction specified in FIP_SBD_EXT.1.5.

Guidance

The evaluator shall verify that the operational guidance provides instructions with how to configure rules using the packet payload string-based detection fields defined in FIP_SBD_EXT.1.2.

The evaluator shall verify that the operational guidance provides instructions with how to configure reactions specified in FIP_SBD_EXT.1.5 for each string-based detection signature.

The evaluator shall verify that the operational guidance provides instructions with how rules are associated with distinct network interfaces that are capable of being associated with signatures.

Tests

The evaluator shall perform the following tests:

- Test FIP_SBD_EXT.1.2:1: The evaluator shall use the instructions in the operational guidance to test that packet payload string-based detection rules can be assigned to the reactions specified in FIP_SBD_EXT.1.5 using the attributes specified in FIP_SBD_EXT.1.2. However it is not required (nor is it feasible) to test all possible strings of protocol data, the evaluator shall ensure that a selection of strings in the requirement is selected to be tested. At a minimum at least one string using each of the following attributes from FIP_SBD_EXT.1.2 should be tested for each protocol. The evaluator shall generate packets that match the string in the rule and observe the corresponding reaction is as configured.
    - Test at least one string of characters for ICMPv4 data: beyond the first 4 bytes of the ICMP header.
    - Test at least one string of characters for ICMPv6 data: beyond the first 4 bytes of the ICMP header
    - TCP data (characters beyond the 20 byte TCP header):
        i. Test at least one FTP (file transfer) command: help, noop, stat, syst, user, abort, acct, allo, appe, cdup, cwd, dele, list, mkd, mode, nlst, pass, pasv, port, pass, quit, rein, rest, retr, rmd, rnfr, rnto, site, smnt, stor, stou, stru, and type.
        ii. HTTP (web) commands and content:
            1. Test both GET and POST commands

        2. Test at least one administrator-defined string to match URLs/URIs and webpage content.
      iii. Test at least one SMTP (email) state: start state, SMTP commands state, mail header state, mail body state, abort state.
      iv. Test at least one string in any additional attribute type defined within the “other types of TCP payload inspection” assignment, if any other types are specified.
    - Test at least one string of UDP data: characters beyond the first 8 bytes of the UDP header;
    - Test at least one string for each additional attribute type defined in the “other types of packet payload inspection” assignment, if any other types are specified.
- Test FIP_SBD_EXT.1.2:2: The evaluator shall repeat Test 1 above to ensure that signature-based IPS policies can be defined for each distinct network interface type capable of applying signatures as supported by the TOE.

FIP_SBD_EXT.1.3

TSS

The evaluator shall verify that the TSS describes how the attacks defined in FIP_SBD_EXT.1.3 are processed by the TOE and what reaction is triggered when these attacks are identified.

Guidance

The evaluator shall verify that the operational guidance provides instructions with configuring rules to identify the attacks defined in FIP_SBD_EXT.1.3 as well as the reactions to these attacks as specified in FIP_SBD_EXT.1.5.

Tests

The evaluator shall create and/or configure rules for each attack signature in FIP_SBD_EXT.1.3. For each attack, the TOE should apply its corresponding signature and enable it to each distinct network interface type capable of applying the signatures. The evaluator shall use packet captures to ensure that the attack traffic is detected by the TOE and a reaction specified in FIP_SBD_EXT.1.5 is triggered and stops the attack. Each attack should be performed one after another so as to ensure that its corresponding signature successfully identified and appropriately reacted to a particular attack.

FIP_SBD_EXT.1.4

TSS

The evaluator shall verify that the TSS describes how the attacks defined in FIP_SBD_EXT.1.4 are processed by the TOE and what reaction is triggered when these attacks are identified.

Guidance

The evaluator shall verify that the operational guidance provides instructions with configuring rules to identify the attacks defined in FIP_SBD_EXT.1.4 as well as the reactions to these attacks as specified in FIP_SBD_EXT.1.5.

Tests

The evaluator shall configure individual signatures for each attack in FIP_SBD_EXT.1.4. For each attack, the TOE should apply its corresponding signature and enable it to each distinct network interface type capable of applying signatures. The evaluator shall use packet captures to ensure that the attack traffic is detected by the TOE and a reaction specified in FIP_SBD_EXT.1.5 is triggered and stops the attack. Each attack should be performed one after another so as to ensure that its corresponding signature successfully identified and appropriately reacted to a particular attack.

FIP_SBD_EXT.1.5

TSS

There are no TSS EAs for this element

Guidance

The guidance EAs for this element are performed in conjunction with FIP_SBD_EXT.1.1, FIP_SBD_EXT.1.3, and FIP_SBD_EXT.1.4.

Tests

The test EAs for this element are performed in conjunction with those for FIP_SBD_EXT.1.1, FIP_SBD_EXT.1.2, FIP_SBD_EXT.1.3, and FIP_SBD_EXT.1.4.

FIP_SBD_EXT.1.6

TSS

There are no TSS EAs for this element.

Guidance

The evaluator shall verify that the operational guidance provides configuration instructions, if needed, to detect payload across multiple packets.

Tests
The evaluator shall repeat one of the tests in Test FIP_SBD_EXT.1.2:1 but generate multiple non-fragmented packets that contain the string in the rule defined. The evaluator shall verify that the malicious traffic is still detected when split across multiple non-fragmented packets.

2.2.3 Security Management (FMT)

FMT_SMF.1/IPS Specification of Management Functions (IPS)

FMT_SMF.1/IPS
TSS
The evaluator shall verify that the TSS describes how the IPS data analysis and reactions can be configured.This may be performed in conjunction with the evaluation of FIP_ABD_EXT.1, FIP_IPB_EXT.1, and FIP_SBD_EXT.1.
Guidance
The evaluator shall verify that the operational guidance describes the instructions for each function defined in the SFR, describes how to configure the IPS data analysis and reactions, including how to set any configurable defaults and how to configure each of the applicable analysis pattern matching methods and reaction modes.
Tests
The evaluator shall perform the following tests:

- Test FMT_SMF.1/IPS:1: The evaluator shall use the operational guidance to create a signature and enable it on an interface. The evaluator shall then generate traffic that would be successfully triggered by the signature. The evaluator should observe the TOE applying the corresponding reaction in the signature.
- Test FMT_SMF.1/IPS:2: The evaluator shall then disable the signature and attempt to regenerate the same traffic and ensure that the TOE allows the traffic to pass with no reaction.
- Test FMT_SMF.1/IPS:3: The evaluator shall use the operational guidance to import signatures and repeat the test conducted in Test 1.

Additionally, for each other claimed management function, the evaluator shall follow the steps listed in the operational guidance for configuration of that function's behavior and verify that configuration has the intended effect (e.g., for modifying the duration of traffic blocking actions, the evaluator could configure this with two separate time values and verify that traffic was blocked for the configured duration in both cases).

Testing for this SFR will generally be performed in conjunction with the test activities for FIP_ABD_EXT.1 and FIP_SBD_EXT.1, as testing various aspects of IPS functionality will necessarily involve this functionality to be configured such that the TSF is in a known state when being tested.

2.3 Evaluation Activities for Optional SFRs

2.3.1 Security Audit (FAU)

FAU_STG.2/IPS Protected Audit Trail Storage

FAU_STG.2/IPS
TSS
The evaluator shall ensure that the TSS identifies how IPS data is protected from unauthorized modification and deletion.
Guidance
The evaluator shall confirm the guidance documentation describes how to protect IPS data from unauthorized modification and deletion.
Tests

The evaluator shall devise tests that demonstrate that IPS data can be protected from unauthorized modification and deletion.

FAU_STG.5/IPS Prevention of Audit Data Loss

FAU_STG.5/IPS
TSS
The evaluator shall ensure that the TSS identifies how IPS data logging is handled once the IPS data trail is full.
Guidance
The evaluator shall confirm the guidance documentation describes any steps involved to manage IPS data logging when the IPS audit trail is full.
Tests
There are no test EAs for this component.

2.3.2 Intrusion Prevention (FIP)

FIP_SBD_EXT.2 Traffic Normalization

FIP_SBD_EXT.2.1
TSS
The evaluator shall verify that the TSS describes how the TOE is able to inspect traffic inside the encapsulation protocols claimed in the requirement.
Guidance
The evaluator shall examine the operational guidance to determine that it contains instructions for inspecting tunneled packets through the encapsulation methods identified in the requirement.
Tests
The evaluator shall set up the conditions necessary to execute testing from FIP_SBD_EXT.1 except that the traffic to be inspected is encapsulated in one of the tunneling protocols that the TOE is capable of inspecting. The evaluator shall transmit the encapsulated traffic and observe that the TSF is able to inspect it and respond in the configured manner. The evaluator shall repeat this test for each type of encapsulated traffic supported by the TSF.
FIP_SBD_EXT.2.2
TSS
The evaluator shall verify that the TSS describes how audit records are generated when packets cannot be reassembled after fragmentation. Also, for inline mode, the evaluator shall examine the TSS to ensure packets are dropped.
Guidance
There are no guidance EAs for this element.
Tests
The evaluator shall perform the following tests:

- Test FIP_SBD_EXT.2.2:1: The evaluator shall generate packets that cannot be reassembled after fragmentation; the evaluator shall ensure audit events are generated for all instances of IP normalization.
- Test FIP_SBD_EXT.2.2:2: For inline mode: The evaluator shall test for automatic packet rejection for when packets cannot be reassembled after fragmentation. The evaluator shall use packet captures to ensure that the IP traffic is detected by the TOE and packets are dropped.
- Test FIP_SBD_EXT.2.2:3: The evaluator shall generate packets that can be reassembled after fragmentation; the evaluator shall ensure audit events are generated for all instances of IP normalization.

FIP_SBD_EXT.2.3
TSS
The evaluator shall verify that the TSS describes that packets are automatically dropped for the following normalization:

- duplicate packets
- changed packets
- out of sequence packets
- other packet types claimed in the requirement, if any

Guidance

There are no guidance EAs for this element.

Tests

The evaluator shall generate the following types of packets and observe that they are discarded by the TSF:

- duplicate packets
- changed packets
- out of sequence packets
- other packet types claimed in the requirement, if any

2.3.3 Protection of the TSF (FPT)

FPT_FLS.1/IPS Failure with Preservation of Secure State

FPT_FLS.1/IPS

TSS

The evaluator shall examine the TSS section to determine that the TOE's implementation of the fail secure functionality is documented. The evaluator shall ensure that the TSS identifies all failures that will result in the TOE preserving a secure state if triggered. The evaluator shall then ensure that the TOE will attain a secure state after inserting each specified failure mode type. The evaluator shall also examine the TSS to determine whether the fail-secure functionality is configurable.

Guidance

There are no operational guidance EAs for this component.

Tests

For each type of failure listed in the assignment, the TOE vendor must provide the evaluator with the means to trigger the failure, and the evaluator shall reproduce each type of failure to ensure that an applied IPS policy remains enforced during the failure. For example, various causes including temporary loss of power could result in a reboot of the TOE. If the active IPS policy at the time of the failure (e.g. reboot) ensured that ICMP echo packets were dropped by the TOE, the evaluator shall confirm that at no point during the shutdown or restart of the TOE is any ICMP echo packet allowed through the TOE (though in this example, it should be understood that there will be a period at which IPS events are not audited while the audit mechanism is pending restart).

2.4 Evaluation Activities for Selection-Based SFRs

The PP-Module does not define any selection-based requirements.

2.5 Evaluation Activities for Objective SFRs

2.5.1 Security Audit (FAU)

FAU_ARP.1 Security Alarms

FAU_ARP.1

TSS

The evaluator shall verify that the TSS includes a description of the alerts specified in the requirement and the events that may trigger these alerts to be generated. The evaluator shall also verify that the TSS states that audit data cannot be transmitted through the security alarms interface.

Guidance
The evaluator shall verify that the guidance explains how to enable alerts specified in the requirement based on the events that the TSS identifies as potential security violations.
Tests
For each action that can be taken by the TSF upon detection of a potential security violation, the evaluator shall configure the TOE to take that action when a particular event occurs. The evaluator shall then perform some action to trigger the event (e.g. by transmitting network traffic that represents the event to the TOE's sensor interfaces) and observe that the configured action occurs in response.

FAU_SAR.1 Audit Review

FAU_SAR.1
TSS
The evaluator shall examine the TSS to verify that it describes the ability of administrators to view IPS data from the IPS events, the format in which this IPS data is displayed, and how an administrator is authorized to view this data.
Guidance
The evaluator shall examine the operational guidance to verify that it provides instructions on how to access and interpret IPS events using the TOE's management interface.
Tests
The evaluator shall devise tests that demonstrate that IPS data (generated as defined in FAU_GEN.1/IPS) can be interpreted by authorized administrators from the TOE's management interface.

FAU_SAR.2 Restricted Audit Review

FAU_SAR.2
TSS
The evaluator shall examine the TSS to determine that it identifies the users that are permitted to view IPS data (e.g., members of a particular group).
Guidance
The evaluator shall examine the operational guidance to determine what actions are needed to grant or revoke a user's ability to view IPS data.
Tests
The evaluator shall log on to the TOE as various users to confirm that only those users that are intended to be granted read access to IPS data are able to view it.

FAU_SAR.3 Selectable Audit Review

FAU_SAR.3
TSS
The evaluator shall verify that the TSS includes a description of how the TOE has the ability to apply filtering and sorting of IPS data using the parameters listed in the requirement.
Guidance
The evaluator shall review the operational guidance to ensure that the guidance itemizes all event types, as well as describes all attributes that are to be selectable in accordance with the requirement, to include those attributes listed in the assignment. The administrative guidance shall also contain instructions on how to set the pre- selection, as well as explain the syntax (if present) for multi-value pre-selection. The administrative guidance shall also identify those audit records that are always recorded, regardless of the selection criteria currently being enforced.
Tests
The evaluator shall perform the following tests:

- Test FAU_SAR.3:1: For each attribute listed in the requirement, the evaluator shall devise a test to show that selecting the attribute causes only audit events with that attribute (or those that are always recorded, as

identified in the administrative guidance) to be recorded.
- Test FAU_SAR.3:2: [conditional]: If the TSF supports specification of more complex audit pre-selection criteria (e.g., multiple attributes, logical expressions using attributes) then the evaluator shall devise tests showing that this capability is correctly implemented. The evaluator shall also, in the test plan, provide a short narrative justifying the set of tests as representative and sufficient to exercise the capability.

2.6 Evaluation Activities for Implementation-dependent SFRs

2.6.1 Resource Utilization

FRU_RSA.1 Maximum Quotas

FRU_RSA.1
TSS
The evaluator shall examine the TSS to ensure that it identifies all resources controlled through the quota mechanism, and that this list contains those resources used to support traffic inspection. The evaluator shall ensure that the TSS describes how each resource is counted as “used” and how a maximum quota or use is determined, as well as the action taken when the quota is reached.
Guidance
The evaluator shall examine the operational guidance to determine that it contains instructions for establishing quotas (if they are configurable) and that it describes any actions administrators can or should take in response to a quota being reached.
Tests
The evaluator shall follow the operational guidance to configure quotas for the resource (if such a capability is provided). The evaluator then causes the resource quota to be reached, and observes that the action specified in the TSS occurs.

3 Evaluation Activities for SARs

The PP-Module does not define any SARs beyond those defined within the base NDcPP to which it must claim conformance. It is important to note that a TOE that is evaluated against the PP-Module is inherently evaluated against this Base-PP as well. The NDcPP includes a number of Evaluation Activities associated with both SFRs and SARs. Additionally, the PP-Module includes a number of SFR-based Evaluation Activities that similarly refine the SARs of the Base-PPs. The evaluation laboratory will evaluate the TOE against the Base-PP and supplement that evaluation with the necessary SFRs that are taken from the PP-Module.

4 Required Supplementary Information

This Supporting Document has no required supplementary information beyond the ST, operational guidance, and testing.

Appendix A - References

Identifier	Title
[NDcPP]	collaborative Protection Profile for Network Devices, Version 4.0, December 22, 2025